

Thynk Unlimited

GRADE 3 CODING: PYTHON TURTLE — DRAW WITH CODE

Let's have fun creating
pictures with code!



WHAT IS PYTHON TURTLE?

Python Turtle is a fun drawing tool that lets you create pictures by writing code! You give commands like move forward or turn right, and the turtle draws lines on the screen. It is like having a tiny robot that follows your instructions!

**Code your imagination —
one turtle step at a time!**



SETTING UP PYTHON TURTLE

Let's Get Started!

First, we import the turtle library and create our turtle friend.

```
import turtle  
t = turtle.Turtle()
```



Type this code in your
Python IDE and press
Run!



BASIC TURTLE COMMANDS

Use these commands to move your turtle and draw on the screen. Try them out and see what happens!

`forward(100)` →
move forward 100
steps

`right(90)` → turn
right 90 degrees

`backward(50)` →
move backward 50
steps

`left(45)` → turn left
45 degrees



DRAWING A SQUARE

```
import turtle  
t = turtle.Turtle()  
for i in range(4):  
    t.forward(100)  
    t.right(90)
```

Let's draw a square by moving the turtle four times. Each side is 100 steps, and we turn right 90 degrees at each corner!

4 sides + 4 turns = a perfect square! 🐢



DRAWING A TRIANGLE

A triangle has three sides and three corners.

Here's the code:

```
import turtle  
t = turtle.Turtle()  
for i in range(3):  
    t.forward(100)  
    t.right(120)
```

Each turn is 120 degrees — that's how we make all three sides equal!

Next, let's make a triangle with three sides.



DRAWING A RECTANGLE

Now, let's draw a rectangle with longer and shorter sides!

```
import turtle  
t = turtle.Turtle()  
for i in range(2):  
    t.forward(200)  
    t.right(90)  
    t.forward(100)  
    t.right(90)
```

A rectangle has two long sides and two short sides. Try changing the numbers!



PEN COLORS & THICKNESS

We can choose any color and make the pen thick or thin. Try different colors to make your drawings bright and beautiful!

`t.pencolor("blue")` — changes the pen color to blue.

`t.pensize(5)` — makes the pen line thicker or thinner.

Try red, green, yellow — any color you like!



PEN UP AND PEN DOWN

Lift the pen to move without drawing. Put it down to draw again. This helps you skip spaces in your picture!

```
t.penup() ← lift the pen  
t.forward(80)  
t.pendown() ← put pen down
```

The turtle moves but leaves a gap — then draws again after `pendown()`!



CHANGING TURTLE SPEED

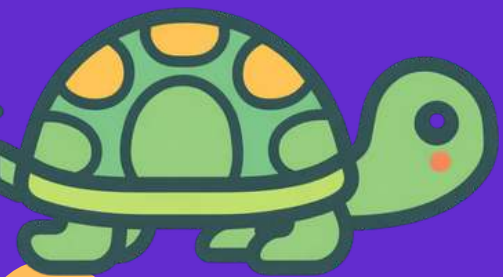
How Fast Does Your Turtle Go?

`t.speed(1)` → Super slow turtle!

`t.speed(5)` → Medium speed

`t.speed(10)` → Zoom zoom, very fast!

Try changing the number to see what happens!



**Make your turtle slow
or fast — you are in
control!**



DRAWING FUN PATTERNS

Let's create cool patterns with loops! Try this code to draw a star:

```
for i in range(5):  
    t.forward(100)  
    t.right(144)
```

Or make a spiral:

```
for i in range(36):  
    t.forward(i * 2)  
    t.right(90)
```

Loops make patterns repeat like magic — try different numbers!



CHALLENGE 1: DRAW YOUR NAME WITH SHAPES!

Use what you learned to draw your name using squares, triangles, and rectangles! Try it out with Python Turtle and see your name come to life with code!

Use squares, triangles, and rectangles to form each letter of your name!

You can be creative and try different colors for each shape!



CHALLENGE 2: CREATE A COLORFUL STAR

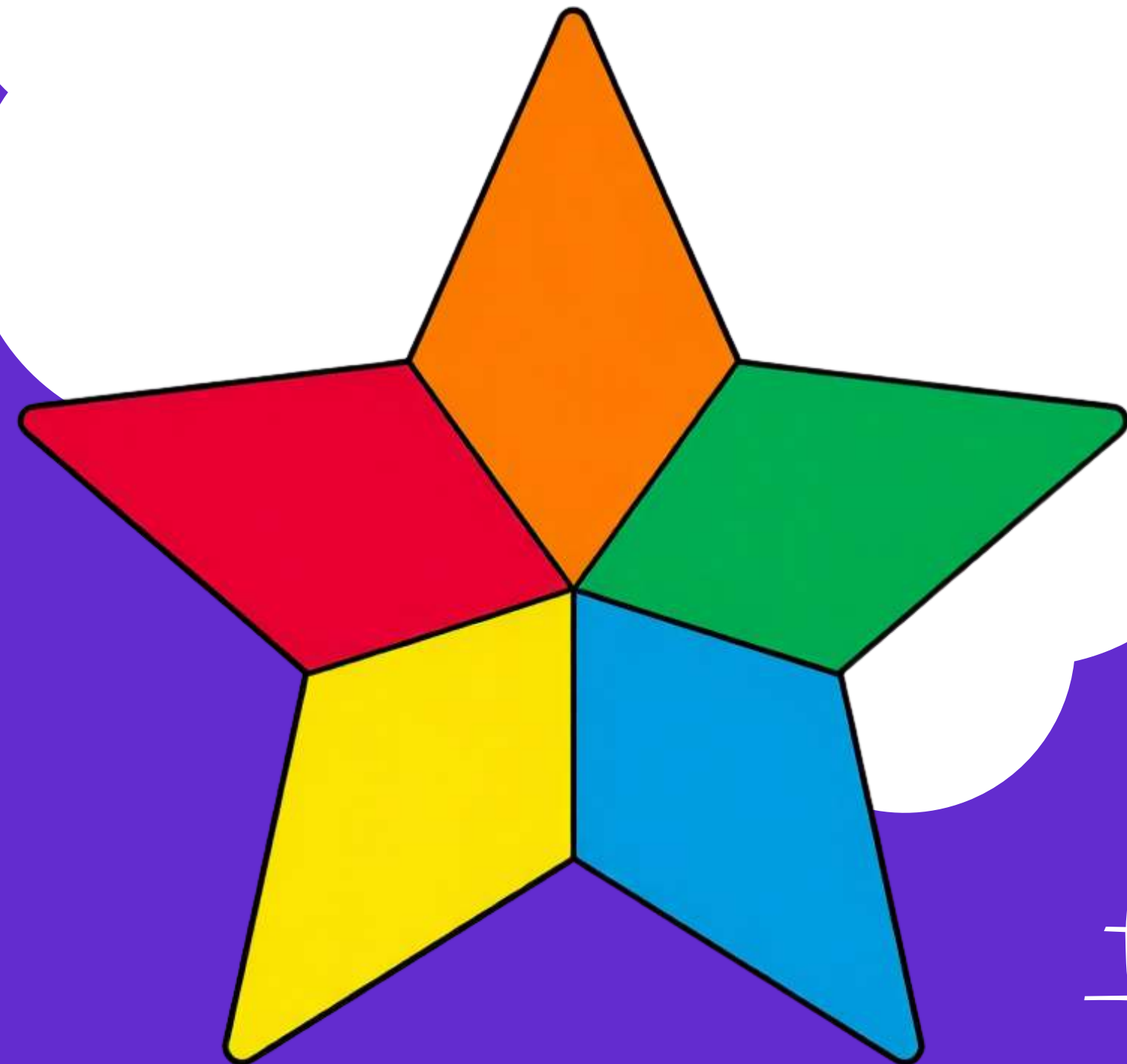
Activity Time!

Try drawing a star with different colors and pen sizes.
Use a loop to repeat the lines and turns!

```
t.pencolor("red")  
t.pensize(3)  
for i in range(5):  
t.forward(100)  
t.right(144)
```



**Hint: A star needs 5
sides — turn right 144
degrees each time!**



CHALLENGE 3: MAKE A PATTERN WITH LOOPS

Use loops to make a repeating pattern like a spiral or flower!

Here's a hint:

```
import turtle  
t = turtle.Turtle()  
for i in range(36):  
    t.forward(100)  
    t.right(170)
```

**How many shapes can
you make with just one
loop? Try it and see!**



SUMMARY: YOU ARE A TURTLE ARTIST!

What We Learned Today

Keep practicing and have fun coding your own art! Every line of code you write makes something beautiful.

- 🐢 Python Turtle intro & basic commands
- 🎨 Shapes, colors & pen control
- ★ Speed, patterns & challenges





THANK YOU AND KEEP CODING!

See You Next Time, Coders!

Thank you for joining Python Turtle class! Keep practicing your drawings and have fun creating art with code. Next session: More shapes, colors, and cool patterns!

**Ready to create more with Python
Turtle? Keep coding and have fun!**